Avishek Bista

Online E-Commerce System for Sales and Order Management

Tribhuvan University

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ABSTRACT

This is an online e-commerce system in terms of sales and order. Management" proposes elaboration and development of the web-based platform. that automates the sales and order taking process and the transactions. The conventional selling approaches are typically found in small and medium businesses. which may lead to manual records keeping or unofficial social media transactions. data inconsistency, inefficient monitoring and ineffectiveness.

The proposed system is a web-based system that provides a centralized solution where customer can. register, log in, shop, work with shopping carts and order. online. Administrators can administer products, viewing and tracking orders. logs of the sales through a special dashboard. The system is developed implementing the modern web technologies on the Django platform on the back end, and data storage Structured data is stored through a relational database.

The methodology to be applied in the development of this project is the Iterative Waterfall that involves the following. implementation, system design, requirement analysis, testing and deployment. planning. It is largely a Transaction Processing System.

(TPS), ensuring that sales and order data must be input correctly and on-demand and providing a point of further development into reporting and analytics. features. The results of the testing have shown that the system is dependable within the key modules. like authentication, managing carts, processing and transaction of orders. recording. The application developed is efficient in terms of operations and data. Validity, and usefulness of e-sales management.

CHAPTER I: INTRODUCTION

1.1 Background of the study

The rapid evolution of technology has reshaped consumer lifestyles and expectations, making digital connectivity a necessity in today's business landscape (Ms. Shipra & Mrs. Colin Janet, 2023). E-commerce has emerged as a powerful platform that simplifies the buying and selling of goods using the internet, offering consumers convenience, cost-effectiveness, and access to a global marketplace. Especially during the COVID-19 pandemic, when physical access to services was limited, digital platforms enabled uninterrupted transactions and contactless delivery. According to the study, E-commerce functions through various technologies like mobile commerce, online transaction processing, and digital marketing, offering a seamless customer experience.Other apps such as Amazon and Uber are not only selling goods but also personalization and interaction, and this enhances the degree of interaction with the consumers. However, security risks like data leaks and phishing are also present as the number of people using them grows, and consumer education and technological protection is essential. This is why one should understand the nature of how e-commerce operates in order to undertake secure, operational, and customer-driven online transactions (IJSREM, 2023).

Electronic commerce (e-commerce) has transformed drastically the way business organizations conduct their purchase and sell activities by enabling the application of electronic tools in running business activities. The fast improvement of the internet usage and mobile technologies have made customers demand the convenience, speed and transparency in purchasing goods. The e-commerce systems also offer an effective way of reaching customers to businesses and more so, the small and medium business or enterprises (SMEs) as it enables them to take orders and maintain records of the sale.

The most common method of manual selling product amongst small retailers or local product manufacturers continue to use the manual sales process or basic social media pages to market products in Nepal. In these old methods there is no proper order tracking, customer management and records of transaction due to which there is inefficiencies, errors and difficulty in tracking the performance of the business. The consistency and accountability of data are also challenging in human process sales and orders.

To overcome these limitations, an automated e-commerce system offers web based system. This system is computerized, it keeps the customer information and orders are made on-demand. The E-Commerce system is a project that aims at automating the sale of its products online via its functionalities which involve registration of users, browsing its products, cart, and ordering its products. The system will be targeted to condensing the management of transactions; reducing the human

error; and creating a platform, where one can make future analytical and managerial decisions.

1.2 Problem Statement

It is a challenge to many small businesses that do not have a centralized digital system that will allow them to process sales and transactions with customers. The traditional methods of selling do not provide sufficient processes of order tracking, cart operations, and transaction history. This means that businesses are likely to lose information, inaccurate documents, slow order placement and delivery, and poor customer service.

Moreover, without automation, the business owners will not be able to analyze the sales performance, the customers or generate reports to make a decision. Lack of systematic structure is also a limitation to scalability, and it would be difficult to expand the operations in terms of increase in the number of customers and transactions. These issues raise the need to have a steady, safe, and automated system of e-commerce that could be able to handle daily transactions and archive legitimate sales records.

1.3 Objectives of the Project

The main objectives of this project are as follows:

i. Create a web-based commerce application for the sale of products.
ii. To allow users to register, log in, and search for and view products.
iii. To allow users to purchase products (via cart and online orders).
iv. Permit accurate sales to be recorded.
v. To permit accurate sales to be recorded.
vi. Permit later reporting and management information functionality.

1.4 Literature Review

1.4.1 Overview of Existing Systems

There are a number of popular e-commerce websites that include Amazon, Daraz, Shopify, and Flipkart, which offer the full scope of online shopping. The systems assist in product management, secure payments, order processing and customer management. Nonetheless, these services can be expensive or too complicated to have a small local business. Due to this, there are several SMEs who choose custom or lightweight e-commerce systems that may be tailored to their as well.

1.4.2 Technologies Used in Similar Systems

The majority of the contemporary e-commerce platforms are designed on the basis of a mixture of front and back-end, database, and deployment technologies to provide scalability, security, as well as efficient handling of transactions.

Table 1.1: Technologies Comparison of Similar E-Commerce Systems

System / Project	Programming Language	Database	Framework	Notes
Amazon (Global E-Commerce Platform)	Java, Python, JavaScript	DynamoDB, Aurora, MySQL	Spring, React	It is built based on a microservices architecture, cloud-native infrastructure (AWS), and advanced recommendation systems
Shopify (E-Commerce SaaS)	Ruby, JavaScript	MySQL	Ruby on Rails, React	offers secure payment and order processing, scalable solutions based on SaaS on e-commerce.
Daraz (Regional E-Commerce Platform)	Java, JavaScript	MySQL	Spring Boot, Vue.js	It is capable of supporting high volume transactions, mobile commerce, and local logistics integration.
Flipkart (Online Retail Platform)	Java, Python	MySQL, Cassandra	Spring, React	t is based on distributed systems and data analytics to process transactions and make it personal.

1.4.3 Research and Academic Works

The information systems and e-commerce platforms are transaction-based, which are extensively researched within the sphere of information systems and financial technologies. Scholars note that the use of the Transaction Processing Systems (TPS) is of significance in documenting, processing, and storing the day to day business transactions in an efficient and precise manner. Organizations depend on such systems to support the operation and are also reliable and consistent in data (West, 2021).

The operation-level information systems (Transaction Processing Systems or TPS) are the core of operational-level information systems of organizations (Laudon and Laudon, 2022). They expound on the fact that TPS undertakes routine and repetitive transactions, such as sales processing, entering orders, billing and recording of payment very quickly and accurately. TPS will eradicate manual work in these basic processes and give solid data capturing, which is essential to day to day business and the organization in the long run.

As pointed out by West (2021), TPS is highly significant when it comes to the accuracy, consistency, and data integrity in the processes of an organization.

According to the study, the superior systems such as the Management Information Systems (MIS) and the Decision Support Systems (DSS) are built on quality data regarding transactions. Their research notes that system generated records minimize discrepancies and time wastage in addition to adopting standard processes. This enhances precision in operations and more trustfulness in digital systems of transactions. The beneficial role of transaction processing in the current information systems is strategy-based as without precise TPS data, it is impossible to develop efficient analytical and managerial systems.

According to Turban et al. (2021), contemporary e-commerce platforms work on an idea that customer interactions, order processing, and recording of transactions are automated based on TPS and in real time. They emphasize in their studies that TPS allows the free flow of activities between customers, sellers and system databases in providing better service fast and better user experience. This automation is necessary to process large quantities of online transactions in an efficient manner.

Campbell, Martinez-Jerez, and Tufano (2022) discovered that automated transaction systems have a considerable reduction in human errors and compliance, in comparison to their manual counterparts. Their study indicates that records that are generated by systems reduce discrepancies and time wastage as well as implement standard procedures. This increases accuracy in operations and increases confidence in digital systems of transactions.

According to Wang (2021) centralisation of digital transaction systems brings about transparency and traceability of organisations. With the help of the structured and centralized records, organizations can trace the transactions more effectively and detect anomalies and improve internal control measures. This level of transparency is of particular concern in such areas as audit preparedness and regulatory compliance.

The article by Bakoben, Giudici, and Figini (2020) demonstrates that when transactional information is structured and collected in digital form, the level of accuracy in predictions and operational planning increases. Their analysis focuses on the fact that good records of transactions enable organizations to study the patterns and trends and hence make better forecasts and make better decisions. These can only be achieved when TPS is implemented in the right way.

According to O'Brien and Marakas (2020), web-based information systems enhance accessibility and operational efficiency, particularly to small and medium-sized enterprises (SMEs). Their article emphasizes that the internet systems enable companies to control their operations without the need to set up expensive systems, as well as to save money on infrastructure expenses. This has rendered web-based TPS solutions very appropriate to the SMEs with small resources.

According to Chaffey and Ellis-Chadwick (2019), e-commerce systems increase the market reach whereby a business is able to operate without limitations as it has geographical limits. Their study elaborates that through digital platforms, organizations are able to reach more customers as it works 24 hours. TPS facilitates this growth with confidence that there is always dependable transaction processing irrespective of time and place.

Kumar and Reinartz (2018) discuss the significance of customer data management in online platforms to ensure increased personalization and customer interaction. They state that the correct histories of transactions enable the businesses to know the preferences and behavior of customers. TPS is crucial towards capturing and storing this information associated with customers.

Heizer, Render, and Munson (2020) clarify that automation in order processing enhances the speed of processing and its accuracy and customer satisfaction. According to their research, automated systems eliminate delays and errors caused by manual system order handling. This makes delivery cycles to be quick and the quality of services better.

According to Le et al. (2023), the essential factors in sustainable growth of digital platforms are scalability of the system and innovation capability. Their analysis gives special emphasis to the fact that transaction systems need to be in a position to deal with the growing data volumes as the business operations grow. Scalable TPS architecture has guaranteed long-term performance and usage of the system.

Bhatt, Chennells and Vause (2023) note that online platforms have to have sound transaction handling systems to ensure currency safety and confidence among users. They say that failure of the system or inaccuracy of the transaction can severely hurt customer confidence. A consistent TPS design is thus a requirement to maintain the online business activity.

According to Laudon (2020), efficient transaction systems enhance the performance of organizations through lowering operation costs and time taken to conduct transactions. Automation helps to reduce the number of redundant operations and the efficiency of the workflow, which enables organizations to allocate resources more efficiently.

Satzger, Psaier, and Hummer (2021) address the topic of modular system design that allows e-commerce companies to adjust to any changes in technology in the future. According to them, TPS architecture that uses modular architecture enables the addition of new features without interrupting the current operations, which will enable the system to evolve over the long term.

Peffers et al. (2018) point out that the design oriented information system projects ought to aim at addressing real business problems by the means of functional system development. Their study contributes to the creation of the useful systems such as e-commerce TPS that directly respond to the operational problems.

According to Jakubik and Teleu (2025), past transaction information in the TPS aids the predictive and analytical models in the future. Proper records of transactions form a basis of advanced analytics, forecasting, and management information system.

According to Zwass (2019), e-commerce systems refer to transaction-based platforms, which are integrated in order to bring together customers, business, and digital processes. The paper presents the fact that TPS is the central mechanism that allows safe and effective interaction between all the involved parties.

Dennis, Wixom, and Roth (2021) emphasize that system reliability and maintainability are enhanced when system development methodologies are provided in a structured manner. A well-designed and documented transaction system makes it stable, and flexible with time.

According to Chen, Chiang, and Storey (2019), transactional data has to be of high quality in the future to support the applications of analytics and business intelligence. TPS guarantees quality of data through the quality and accurate records of transaction at the level of operation.

As concluded by Alter (2022), the strategic value of well-designed transaction processing systems is the long-term benefit to the organization by helping organizations to operate effectively and strategically. TPS is a highly effective bridge between the day-to-day activities and the top-level management system.

On the whole, literature reviewed presents the progress in the area of e-commerce and financial systems development as a Transaction Processing System, which can be further improved to become a Management Information System (MIS) by adding reporting, analytics, and decision-support functions.

1.5 Development Methodology

The project follows the Iterative Waterfall Model as its development methodology. The model is a blend of the defined stages of the conventional waterfall method requirement analysis, system design, implementation, testing, and deployment and cyclic feedback and improvement. Every phase is done in systematic order and with some possibilities of making improvements in accordance with testing and user reviews.

This approach will make sure that the e-commerce system is created in a systematic and coordinated way. The ability to repeat incurs that problems detected in the development can be rectified early to give a stable and reliable application. The method is applicable in this project because it would bring clarity in the requirements and incremental addition of the features.

Figure 1.1: The E-Commerce System iterative waterfall model.

1.6 Scope and limitation of the project

The E-Commerce System would be developed with the aim of supporting the online sale with the system anticipated to support centralization of products, users, carts and even orders. It will allow the customers to view all the products, have the opportunity to shop in the cart and purchase the products online. The entire transactions are documented into the system hence qualifies to be a Transaction Processing System (TPS). The project as well offers future extensions of the system to sales reports, analytics dashboards and inventory tracking that can enhance the system into a Management Information System (MIS).

The system is most applicable to the small and mid-size businesses, to the local manufacturers and start-ups who are interested on an easy and straightforward online platform of sales.

Limitations

The E-Commerce System has weaknesses that are:

i. The system would need good internet connection since it is web based.
ii. It could inhibit or imitate the existence of online payment gateway.
iii. The current one does not have sophisticated analytics and reporting.
iv. There is no full use of inventory management and supplier modules.

1.7 Report Organization

This part of the report gives one an impression of how the BIM Minor Project Report is designed.

Table 1.2: Report Structure Overview

Chapter No.	Chapter Title	Description
Chapter I	**Introduction**	Provides background, problem statement, objectives, related works, methodology, scope, and report organization.

Chapter II	**System Development Process**	Discusses system analysis, design, requirements, feasibility, and system architecture.
Chapter III	**Conclusion and Recommendations**	Explains tools, frameworks, and the development process with coding and database design.
References		Lists all sources, journals, and materials referred to.
Appendices		Includes screenshots, code snippets, and supporting documents

CHAPTER II: SYSTEM DEVELOPMENT PROCESS

2.1 Analysis

The E-Commerce System analysis stage has been guided towards determining the weaknesses of the traditional selling and order processing processes and how these activities can be computerized by the use of a web-based system. Most of the small and medium-sized businesses use manual sales tracking, social media messages, or unstructured data and is likely to provide inaccurate order data, ineffective customer management and inability to monitor the transactions.

To overcome these challenges, the requirement analysis has been done by carrying out research on the available e-commerce platforms, expectations and assessment on user needs and manner in which the transactions are being undertaken. The analysis indicated that there was the necessity to possess system that will be able to handle users, products, carts and orders in an automated way. The non-functional and functional requirements were based on the findings that were translated to the system design and implementation.

2.1.1 Requirement Analysis

Requirement analysis was to be introduced in order to fill the gap between the traditional modalities of selling and the suggested automated e-commerce. The most important features were user identification that was secure, browsing the product, carting, order placement and recording of the transactions. The other critical system requirements that were proved to be critical were the usability, performance and security.

2.1.1.1 Functional Requirements

The capabilities of the system will be the following:

Table 2.1: Functional Requirements

Requirement ID	Requirement Description	Example / Details
FR1	User Authentication & Authorization	Users must register and log in to access system features.
FR2	Product Management	Show products (name, price, description, and pictures).
FR3	Cart Management	Users are able to add, change or delete items in the cart.
FR4	Order Handling	Take orders and archive order information.
FR5	Transaction Recording	Recording of all sales transactions.
FR6	Order History & Status	Past orders and order status may be viewed by the user.

FR7	Admin Dashboard	Admin will be able to administer products and see all the orders.

2.1.1.2 Use Case Scenario

The case scenario is whereby the system would be described on how to communicate with the users in an attempt to accomplish a set objective. The use cases of the system include the following and discussed in the following tables:

Table 2.2: Use Case – User Login

Use Case	**User Login**
Actors	User and Admin
Preconditions	User should be registered.
Steps	1. Click Login. 2. Enter email and password. 3. System validates credential. 4. Redirect to dashboard.
Postcondition	User is logged in or error message is displayed.

Table 2.3: Use Case – Add Product to Cart

Use Case	**Add to Cart**
Actors	User
Preconditions	User may or may not be logged in.
Steps	1. Select products. 2. Choose quantity. 3. Click “Add to Cart”.
Postcondition	Product will be added to cart.

Table 2.4: Use Case – Place Order

Use Case	**Place Order**
Actors	User
Preconditions	Products are in cart.
Steps	1. Review Cart. 2. Enter delivery details. 3. Confirm order.
Postcondition	Order successfully placed.

Table 2.5: Use Case – Generate Report

Use Case	**Generate Sales Report**
Actors	Admin
Preconditions	Admin logged in.
Steps	1. Select report type. 2. System generates report.
Postcondition	Sales report displayed.

2.1.1.3 User Roles & Permissions

The system defines different types of users with specific roles and access levels to ensure security and efficient workflow:

Table 2.6: User Roles and Permission

Role	Permissions
Admin	Manage products, view all orders, generate reports
User	Browse products, manage cart and place orders
System	Processes transactions and updates order records

2.1.1.4 Non-Functional Requirements

Non-functional requirements are those requirements that outline how the system is able to function, its limitations, and how it is meant to work with the emphasis given to the quality aspects of the system such as performance, usability, compatibility, and many others. Such requirements specify the manner in which the system is supposed to be done. The non-functional requirements of online order management system are represented below:

Table 2.7: Non-Functional Requirements Table

Requirement ID	Requirement	Description
NFR1	Usability	Simple and intuitive interface
NFR2	Security	Secure login and protected data
NFR3	Performance	Fast page load and transaction processing
NFR4	Scalability	Supports increasing users and orders
NFR5	Compatibility	Works on modern browsers and mobile devices
NFR6	Maintainability	Modular and easy to update

2.1.2 Feasibility Study

i. **Technical Feasibility:** Uses web technologies (HTML, CSS, JavaScript) and backend frameworks are used based on e-commerce.
ii. **Economic Feasibility:** Open-source tools reduce the development cost.
iii. **Operational Feasibility:** Simpler interface that is appropriate in small businesses.
iv. **Legal Feasibility:** Complies with basic data protection standards.

v. **Schedule Feasibility:** Feasible schedule with the help of the Iterative Waterfall Model.

2.1.3 System Requirements Specification (SRS)

Below are minimum and ideal hardware and software requirements for optimal performance, scalability, and security.

Table 2.8: Hardware Requirements

Component	Minimum	Recommended
Processor	Intel Core i3	Intel Core i5
RAM	4 GB	8 GB
Storage	128 GB	256 GB
Internet	5 Mbps	10 Mbps

Table 2.9: Software Requirements

Component	Specification
Operating System	Windows / Linux
Frontend	HTML, CSS, JavaScript
Backend	Django
Database	SQLite (development), PostgreSQL (deployment)
Browser	Chrome, Firefox, Edge

2.2 System Design

System design is a very important step in the E-Commerce System development as it converts the system requirements into a blue print on how to implement the system. This phase determines the general architecture, database design, user interface design, system workflow and object-oriented design. System design aims at ensuring that the application is efficient, scalable, secure, and user-friendly and also capable of processing the transactions correctly.

The e-commerce system design is aimed at the daily sales transactions management, users and products management, and storage of order-related information in a central database. The system is meant to operate as a Transaction Processing System (TPS) and can in future extend to a Management Information System (MIS).

2.2.1 System Architecture

The E-Commerce System has been developed in a three tiers client-server model that facilitates the separation of the application into presentation, application logic and data layers. The architecture enhances maintainability, scalability and security. The system consists of:

i. **Frontend:** Browsing and product purchase interface.
ii. **Backend:** Handles business logic and transaction.
iii. **Database:** Archives user, product and order data.

This multi-layered architecture provides ease of communication between users and the system and also ensures the data integrity in the process of transaction.

2.2.2 Database Design

Database design determines the way the data are arranged, stored, and retrieved in the E-Commerce System. An effective database will guarantee uniformity of data, minimize redundancy, and facilitate effective processing of transactions. As the system deals with high frequency transactions related to orders, the database is vital in the reliability and accuracy of the system.

The database contains user information, product information, cart information, order information and transaction information. The relationships among entities are aimed at quick access to the order history, user activity, and product information.

2.2.2.1 Database Schema

Database schema is the logical view of the database, comprising of tables, attributes and relations..

Table 2.10: Database Scheme

Table Name	Attributes	Description	Relationship
Users	user_id, name, email, password, role	Stores registered user and admin information	Primary table
Products	product_id, name, price, description, stock	Stores product details available for sale	Referenced by Cart and Orders
Cart	cart_id, user_id, product_id, quantity	Temporarily selected items	Linked with Users and Products
Orders	order_id, user_id, order_date, total_amount, status	Stores order information	Linked with Users
Order_Items	item_id, order_id, product_id, quantity, price	Stores individual items of each order	Linked with Orders and Products
Transactions	transaction_id, order_id, payment_mode, date	Stores transaction records	Linked with Orders

Such database structure allows the system to store and access transaction data efficiently thus it is appropriate in TPS operations.

Entity Relationship Diagram (ERD) is the logical design of the database of Online E-Commerce System. It shows relations between the central objects like User, Product, Cart, Order, Order_Item, and transaction. The ERD guarantees data consistency and effective processing of transactions since it clearly specifies primary keys, foreign keys and entity relationships. This structure allows the sales transactions, history of orders and customers activities to be accurately recorded and hence the system is fit to be used in the operation of the Transaction Processing System (TPS).

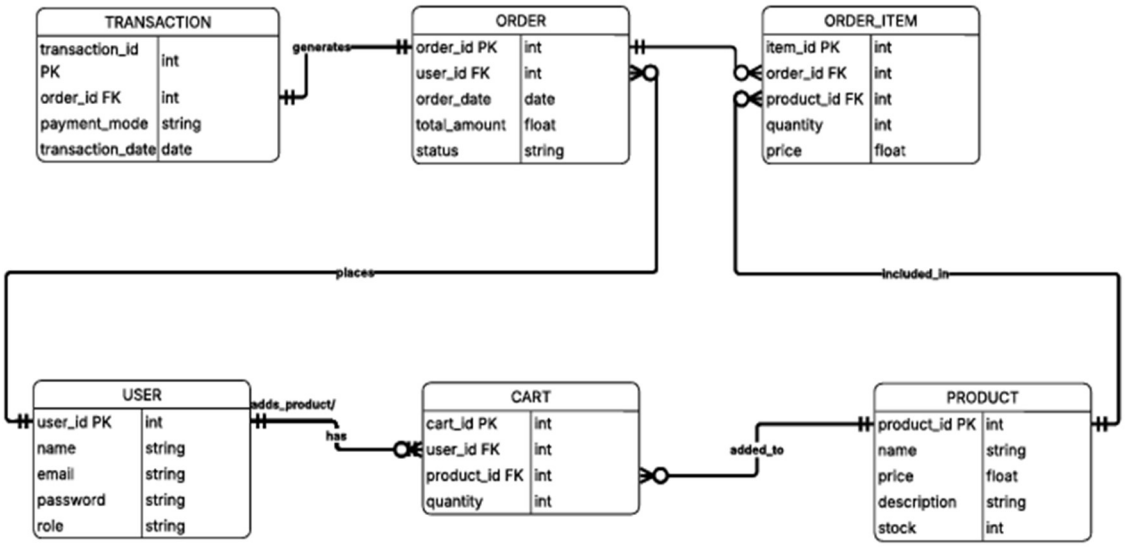

Figure 2.1: Entity Relationship Diagram

2.2.3 User Interface Design

The User Interface (UI) the E-Commerce System has been designed to offer easy, user-friendly, and responsive experience to users and administrators. The design focuses on the simplicity of navigation, simplicity of information, and little effort to use in order to accomplish transactions.

The UI is developed in accordance with the principles of usability to make it possible to navigate products, operate carts, and order items without challenges even by the non-technical users.

2.2.3.1 Major UI Components

Dashboard: The dashboard is the view that gives an overview of an activity in the system.

i. Admin Dashboard: Shows the total products, total orders and the summary of sales.
ii. User Dashboard: Shows recent orders and account information.

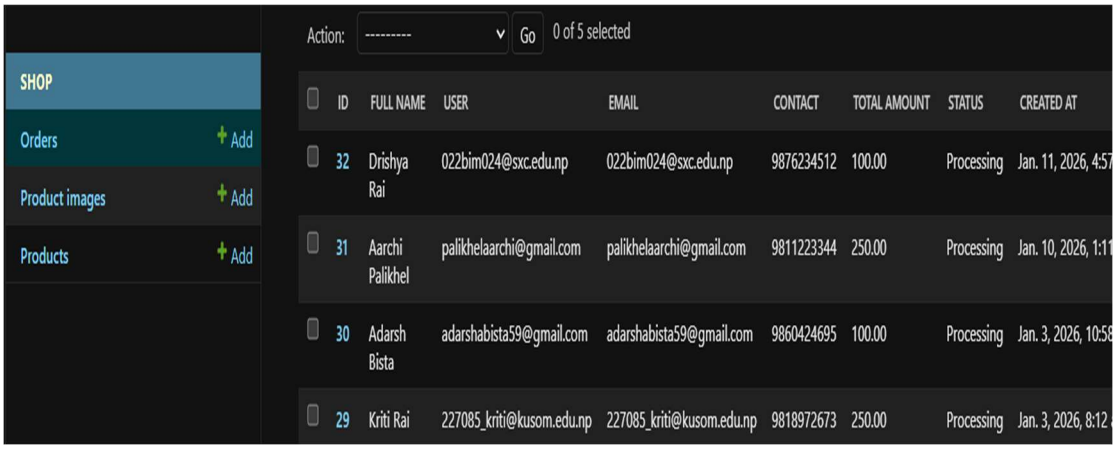

Figure 2.2: Admin Dashboard

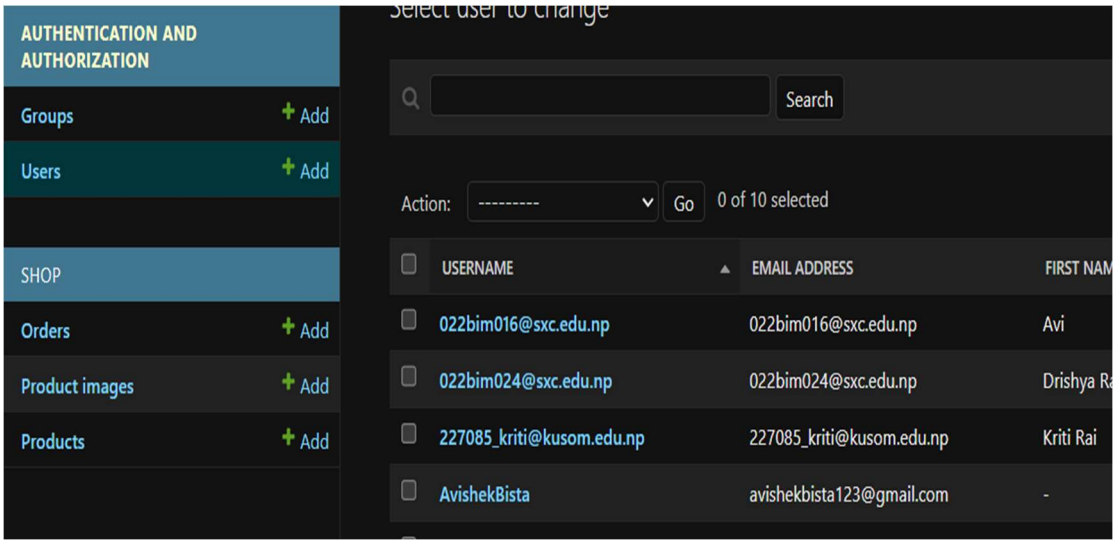

Figure 2.3: User Dashboard

Pages in the System

i. Login & Signup Page: This is where users are allowed to create an account and log in.

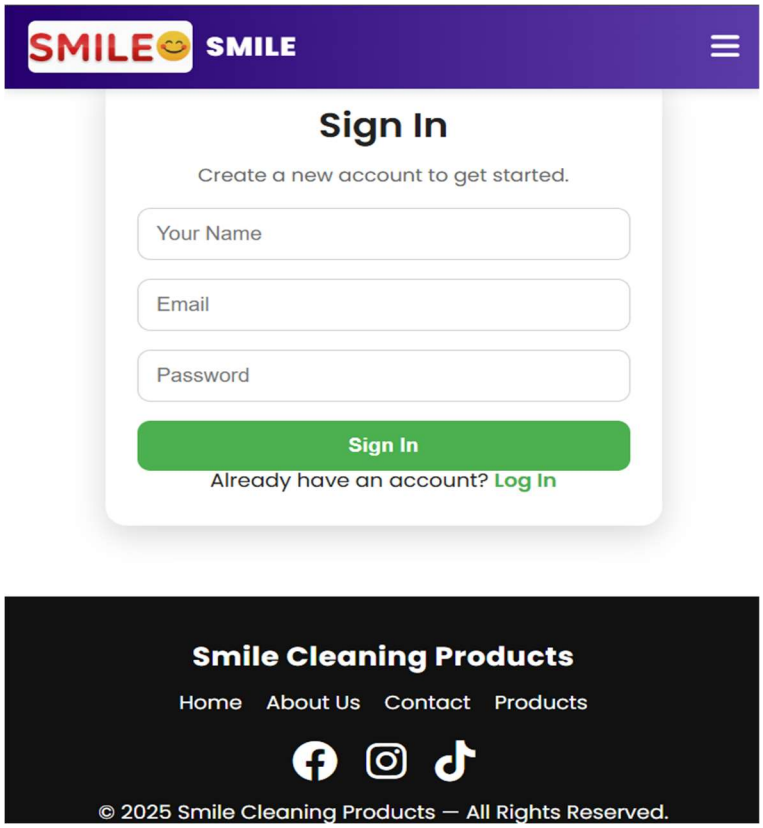

Figure 2.4: Sign In Page

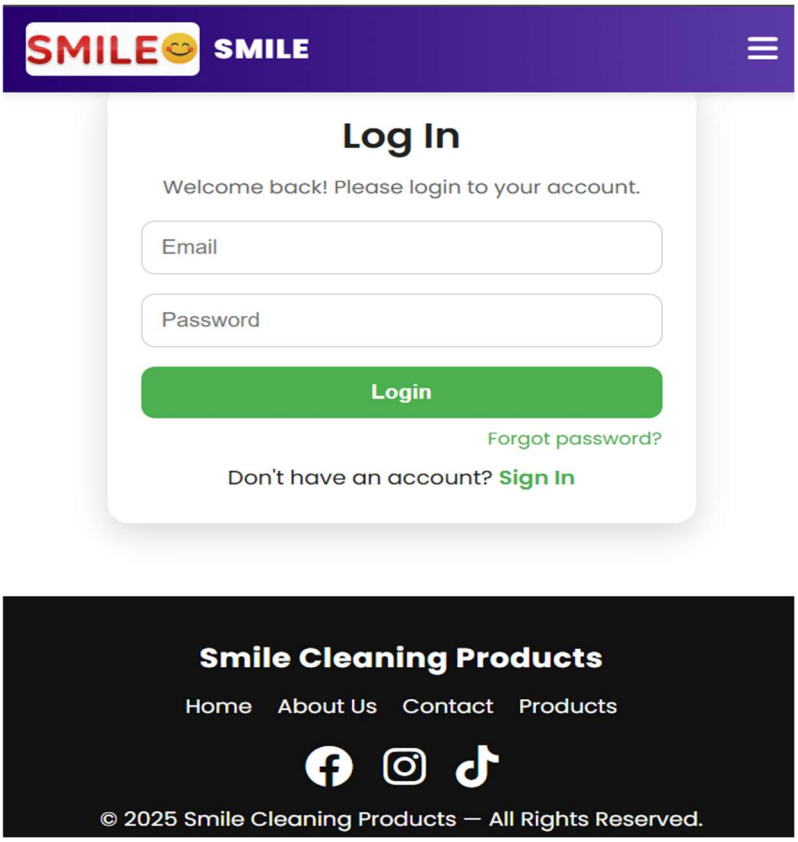

Figure 2.5: Log In Page

ii. Home Page: Displays promoted and offered products.

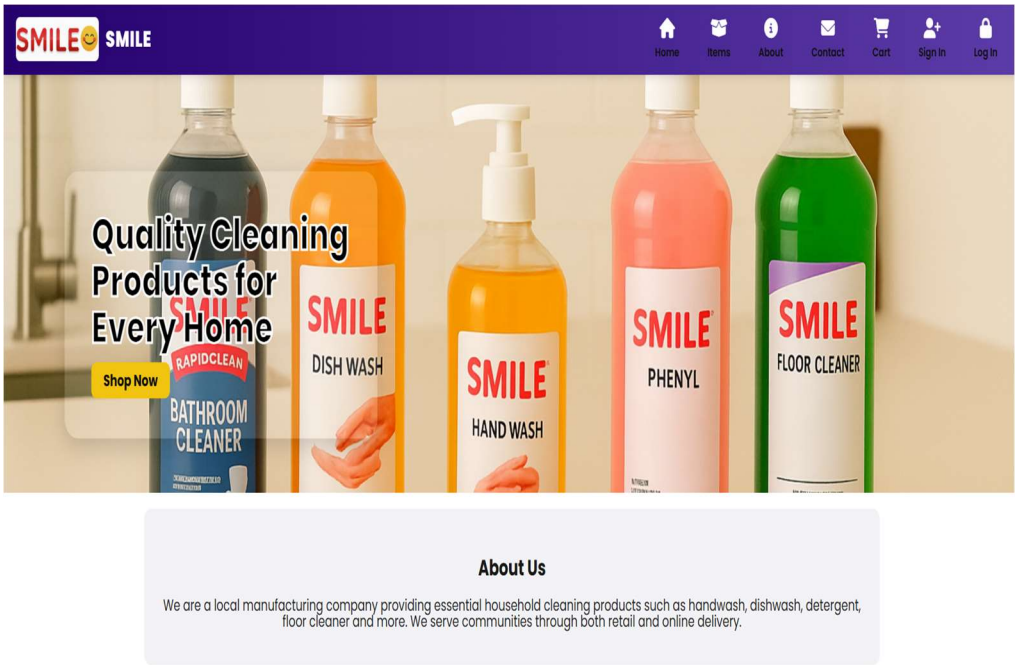

Figure 2.6: Home Page

Product Listing Page:Shows the entire product listing in price and description.

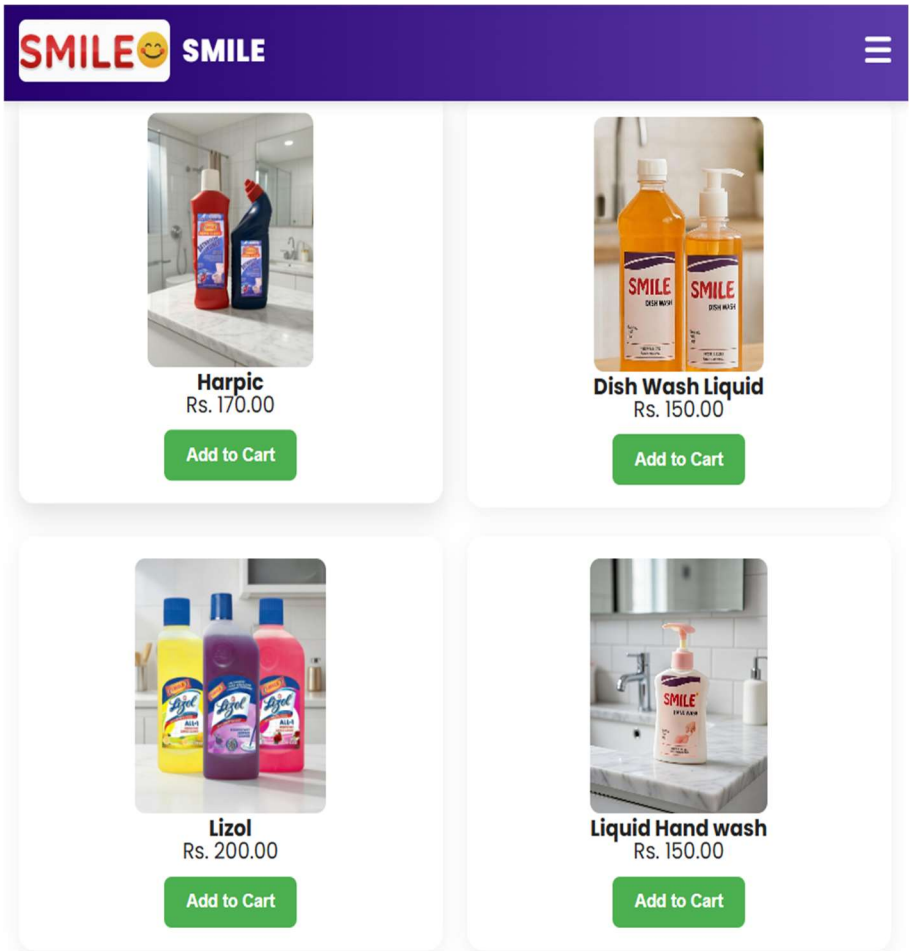

Figure 2.7: Product listing Page

iii. Product Detail Page: Shows detailed information of a product of choice.

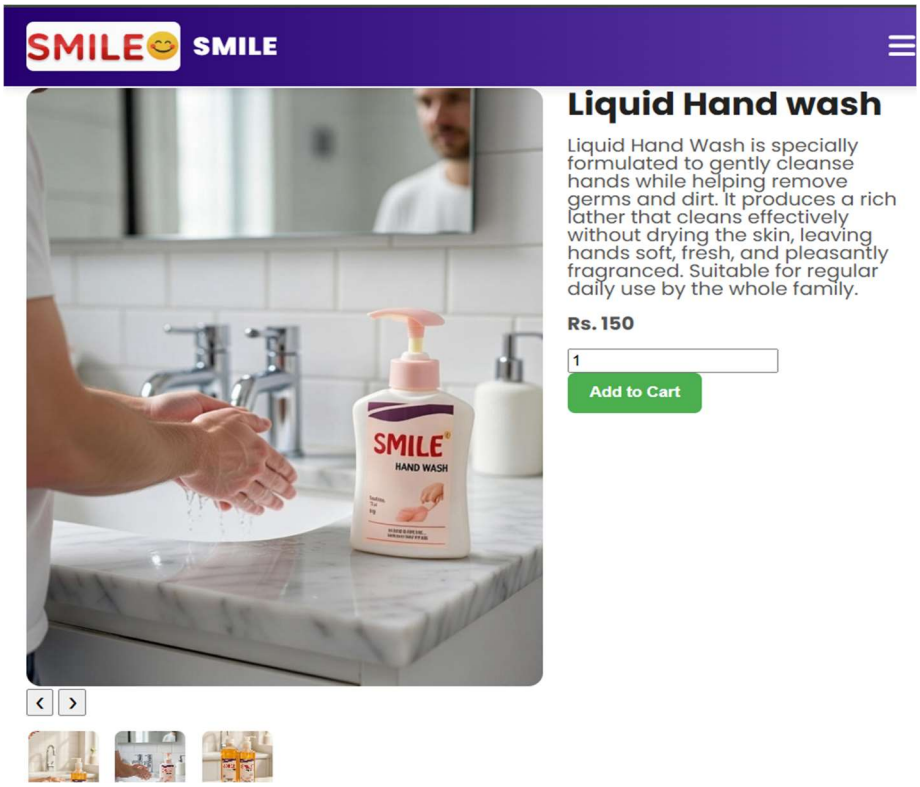

Figure 2.8: Product Detail Page

iv. Cart Page: Shows selected items and total cost.

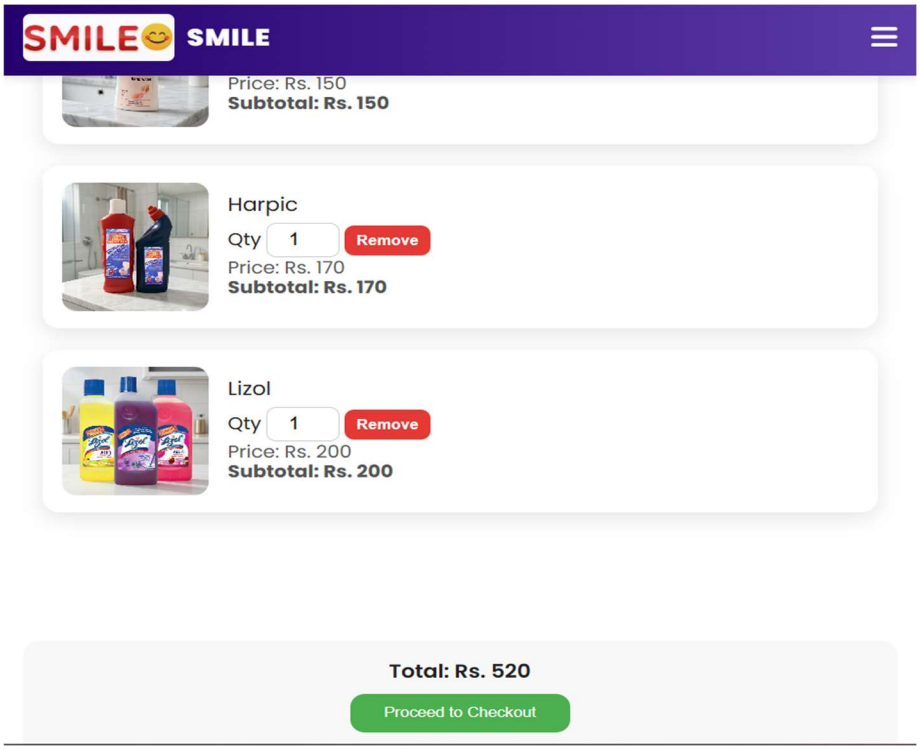

Figure 2.9: Cart Page

v. Checkout Page: This allows confirmation of the items ordered via the website.

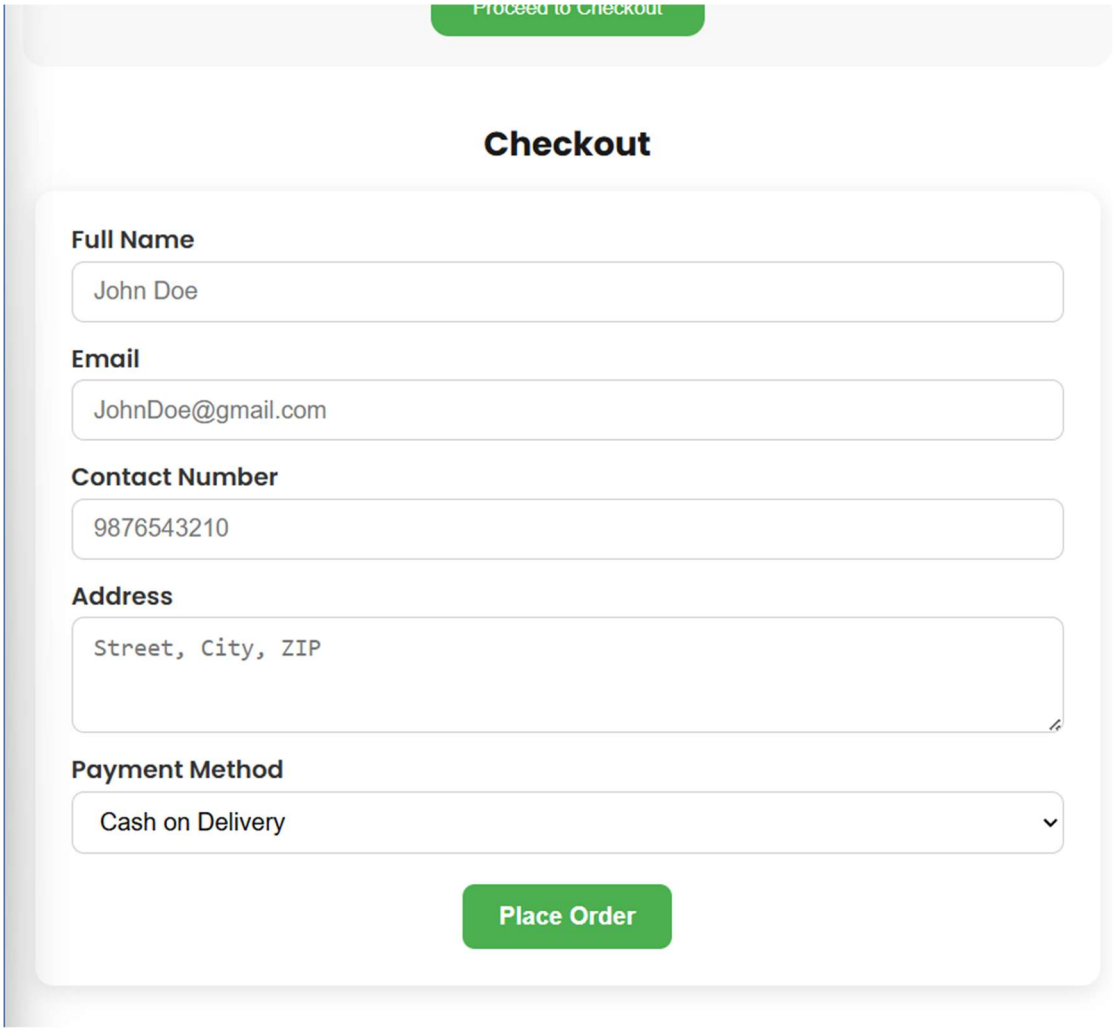

Figure 2.10: Checkout Page

2.2.4 System Flowchart

A system flow chart is an illustration of the rational flow of events in the E-Commerce System. It shows the chronology of the process of fulfilling a transaction, including user logins to order placements.The flowchart includes:

i. User authentication
ii. Product selection
iii. Cart management
iv. Order confirmation
v. Transaction recording
vi. Order completion

The flowchart will assist in visualizing the movement of the data within the system, as well as the decisions that are made at any given stage.

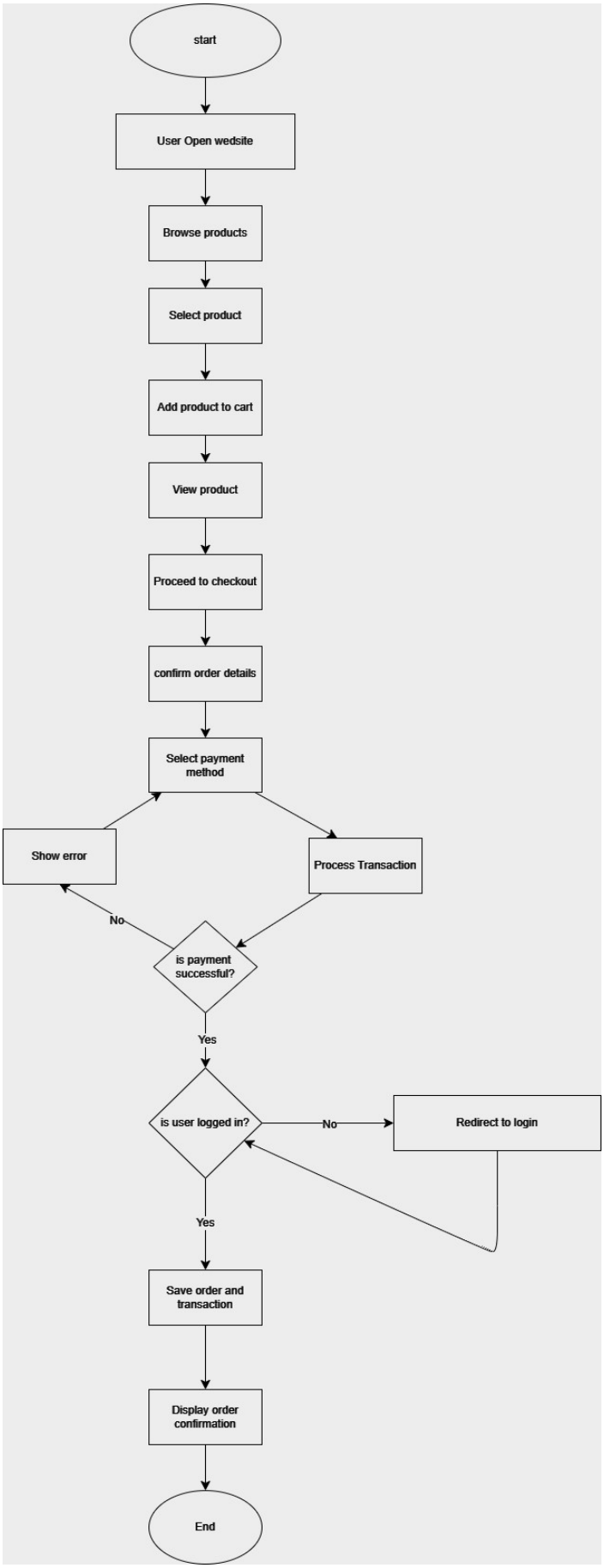

Figure 2.11: Flowchart of Online E-Commerce System

2.2.5 Object-Oriented Modelling

Object-oriented modeling is used to represent the system using objects, classes, and relationships. This approach improves modularity, reusability, and maintainability of the system.

2.2.5.1 System Overview

The E-Commerce System is the application that is based on the web and is aimed to deal with purchasing and selling operations online. It allows users to shop, make orders and track purchase and salesmen manage products and sales.

The system is based on object-oriented principles that arrange the functionalities into reusable units like User, Product, Cart, Order and Transaction classes. This architecture is scaled and feature-enhanced easily.

2.2.5.2 Use Case Model

A case model gives a top-level performance of the system as it includes interactions between the user and the system.

Primary Actors:

i. User
ii. Admin

Major Use Cases:

i. Register / Login
ii. Browse Products
iii. Add to Cart
iv. Place Order
v. View Order History
vi. Manage Products (Admin)
vii. Generate Reports (Admin)

The case model ensures that all functional requirements identified during analysis are properly represented and implemented in the system.

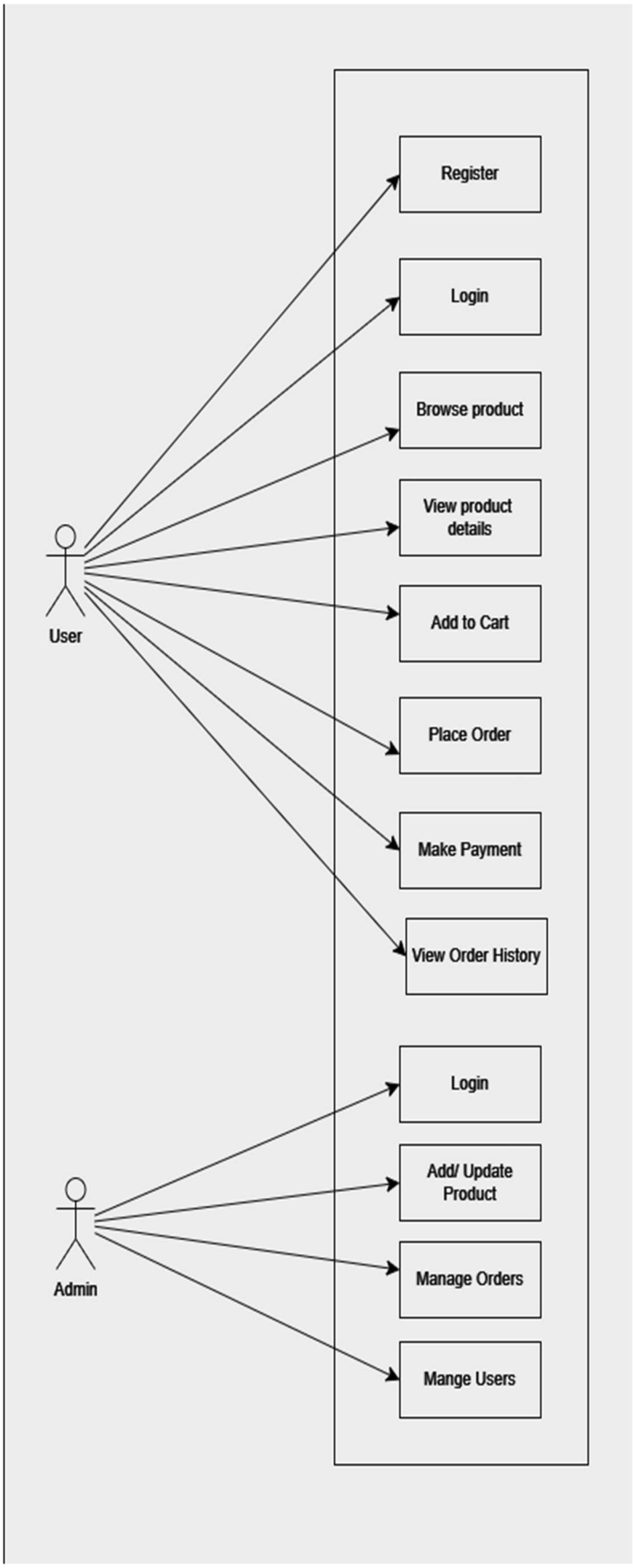

Figure 2.12: Use Case Diagram of Online E-Commerce System

2.2.5.3 Class Diagram

The Class Diagram represents static structure of the Online E-Commerce System using object-oriented principle.

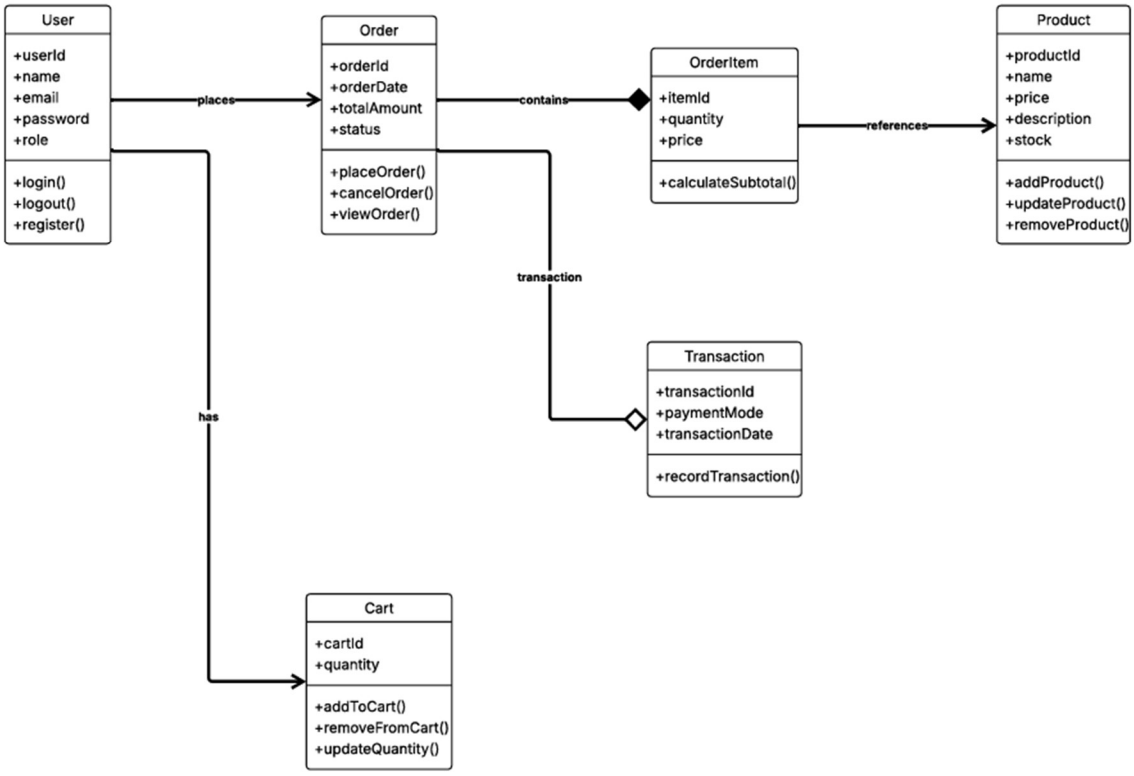

Figure 2.13: Class Diagram of Online E-Commerce System

2.2.5.4 Object Diagram

Object Diagram depicts a runtime image of the Online E-Commerce System. It is an actual object that is formed out of the classes in the course of system running. This flow chart shows the way a user chooses a product and puts it in a cart, orders it and finalizes a transaction.

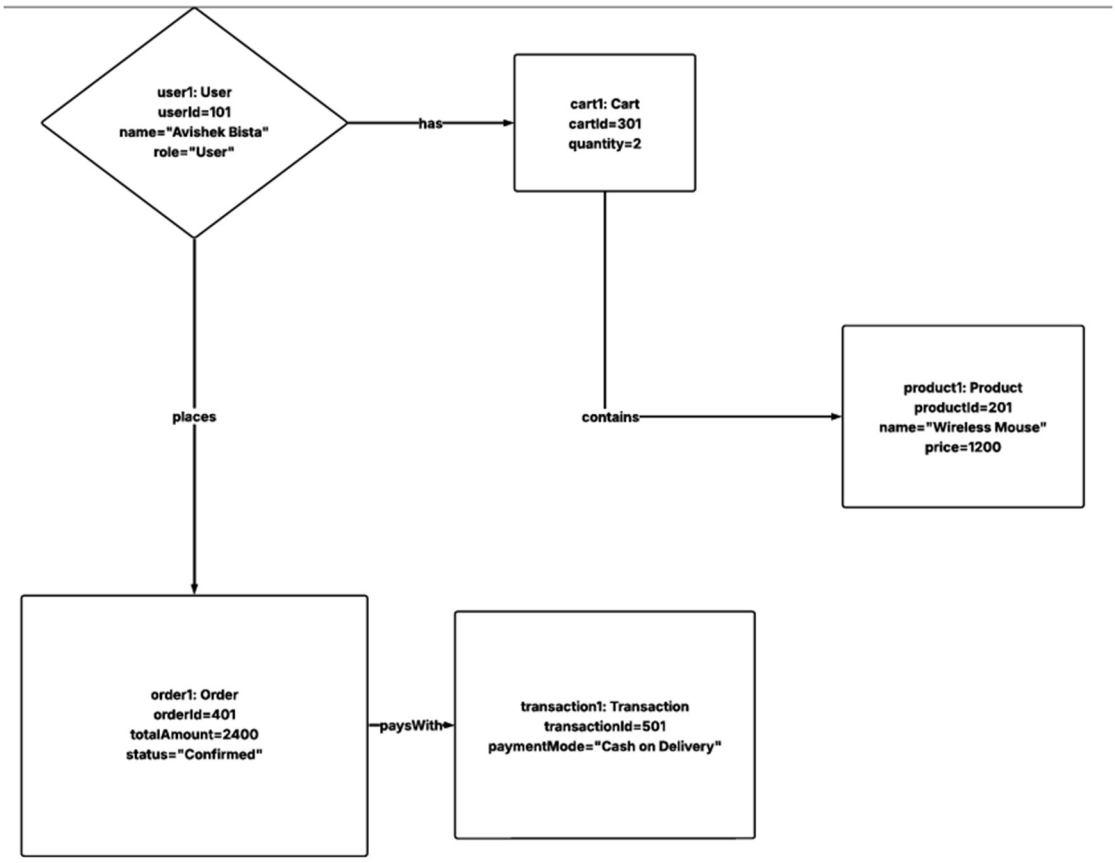

Figure 2.14: Object Diagram of Online E-Commerce System

2.3 Implementation

It implemented the system as the web-based application, which involves the frontend and the backend by using RESTful APIs. Secure authentication, proper order processing and easy user interaction were all part of its implementation.

2.3.1 Tools and Development Environment

Table 2.11: Tools and Development Environment

Tool	Category	Purpose
HTML/CSS	Frontend	Frontend Structuring and styling of the pages.
JavaScript	Logic	Java scripting Logic Dynamic interaction.
Django (Python)	Backend	Server-side/Backend processing

SQLite (development)	Database	Database data storage
VS Code	IDE	Development

2.3.2 System Modules and their Implementation

The E-Commerce System consists of a number of functional modules, which collaborate in facilitating the online purchasing and selling processes. The modules are both independent but also interconnected and therefore achieve scalability, maintainability and high transaction processing. The modular architecture will enable the system to operate as a Transaction Processing System (TPS) as well as to be expanded to a Management Information System (MIS) in future.

Table 2.12: System Modules and their description

Module	Description
User Authentication Module	Manages user registration, login, logout, and session handling. Users must be authenticated before placing orders. The system differentiates between users and administrators, granting role-based access to dashboards and functionalities.
Product Management Module	Enables the administrators to add, delete and update products. Product information is dynamically stored and shown to the users in the form of name, price, description, stock quantity, and images.
Cart Management Module	Allows the user to add or take-out products to a shopping cart in order to update the quantities. The cart data are recorded on a case by case basis so that the user is given the continuity to the checkout.
Order Management Module	Deals with order placement, order confirmation and order status tracking. Every order is associated with a particular user and placed in the database as transaction history.
Transaction Module	All purchase transactions (including the mode of payment, amount of order and date of transaction) are recorded. All the transactions are time stamped to facilitate proper transaction processing and reporting.
Admin Management Module	Allows the management to manage users, products and orders. Admins are able to observe the activity of the system and provide the efficient business work.
Dashboard Module	Provides a summary of the information like the number of products, number of orders, and the latest transactions. The admin dashboard is helpful in keeping track of the performance of the system, whereas users are allowed to see the order history and account information.

All modules are connected with the centralized database to provide the correct storage and retrieval of data in the database that assist in real-time processing of transactions.

2.3.3 Coding Standards and Conventions

i. In order to retain the code quality, readability and scalability, uniform coding practices were observed during the design of E-Commerce System.

ii. Naming Conventions

- Variables and functions follows camelCase
- Model and name of classes follows PascalCase
- The naming of files is lowercase or PascalCase based on the usage

iii. File Organization

The project follows a modular structure separating frontend and backend logic.

iv. Documentation

- The naming of functions and modules was done in a descriptive manner.

v. Best Practices

- There was consistent indentation and formatting.
- Redundancy was reduced through the use of reusable components and functions.
- There was input validation at the frontend and at the backend.
- Configuration data that was sensitive was separated out of core logic.

2.3.4 Security Considerations

The design and development of the E-Commerce System were done with a lot of security in mind to safeguard the user data and transactional information.

i. **User Authentication Security:** User login credential is strongly managed and authenticated user can only make an order or access sensitive functionalities.

ii. **Input Validation:** All user inputs e.g. login credentials, order quantities and form input are all validated to eliminate invalid or malicious entries.

iii. **Access Control**: The administrative operation like product management and order monitoring will be limited to authorized users.

iv. **Secure Deployment (Future Scope):** In future deployment the system will be deployed through HTTPS to provide the secured communication between the client and the server.

Such steps guarantee confidentiality, integrity and system data availability.

2.3.5 Testing and Debugging

The testing was done at various levels to provide the accuracy, reliability and stability of the E-Commerce System

2.3.5.1 Unit Testing

Each individual module was tested separately, which included the authentication of the user in the case of login, updates of the cart, and placement of an order.

2.3.5.2 Integration Testing

Integration testing verified smooth interaction between modules, such as cart data being correctly transferred to the order and transaction modules during checkout.

2.3.5.3 System Testing

The complete workflow from user login to order confirmation was tested to ensure the system functions as intended.

2.3.5.4 User Acceptance Testing (UAT)

Users and administrators tested the system under realistic conditions.

2.4 Deployment

The E-Commerce System was developed and tested in a local environment before planning for online deployment.

2.4.1 Local Deployment Configuration

i. **Frontend:** HTML, CSS, JavaScript rendered via Django(python) templates.
ii. **Backend:** Django development server (python manage.py runserver)
iii. **Database:** SQLite

2.4.2 Planned Online Deployment

i. **Backend:** For backend I deployed my project at a free hosting site called PythonAnyWhere and works in my laptop using Django development server (python manage.py runserver)
ii. **Database:** SQLite
iii. **Static Files:** Served via cloud storage
iv. **Security:** HTTPS was enabled with the use of the SSL certificates.

CHAPTER III: CONCLUSION AND RECOMMENDATIONS

3.1 Summary

This project was named Online E-Commerce System in Sales and Order Management, which was created to solve the problems of the small and medium size business to manage sales, orders and records of transactions in the traditional or manual manner. In Nepal, a lot of local vendors still use informal methods of record keeping, including handwriting or social media communication, which lead to inconsistency of data, lack of order tracking and business information.

The suggested system was developed and deployed as an online e-commerce platform that automates the most crucial business processes such as user registration and authentication, product browsing, cart management, order placement, and recording transactions. The system is developed according to the Iterative Waterfall Development Model that provides a structured but flexible development.

The app was developed with the latest web technologies and the frontend, backend, and database layers are distinguished. This system has been mainly used as an Application System or a Transaction Processing System (TPS) to take daily record of sales transactions and order transactions with accuracy and security. It also offers a good basis upon which it can be enhanced in the future to become a Management Information System (MIS) so that it can store the data appropriately to be used in reporting and analytics.

Unit, integration, system, and user acceptance tests were done to ensure that the system satisfies its functional and non-functional specifications. On the whole, the project has managed to show how automated e-commerce system can enhance efficiency, accuracy, and transparency in sales and order management.

3.2 Conclusion

The Online E-Commerce System of Sales and Order Management meets the main goal to offer the reliable, structured and user-friendly system of online sales transactions management. The project solves some of the major issues that include inaccurate record-keeping, slow order processing, and absence of a record of transaction by changing manual and semi-digital practices with integrated web-based system.

The system is able to support key e-commerce activities such as user authorization, product management, cart, order processing and recording transactions. Its modular design makes it maintainable and scalable, also suitable to a small and medium-sized enterprise looking to get a cost-effective digital solution.

Academically and practically, the project shows how the concepts of the Transaction Processing System (TPS) were applied in a real life situation. It specifies the importance of proper and reliable data on transactions as the working core of any business and the basis of future decision-support and management systems.

To sum up, the project is able to fulfill the stated goals and demonstrate that the properly designed e-commerce system can positively influence the functioning of business, customer experience, and reliability of data.

3.3 Recommendations

Despite the fact that the present version of the system serves the purpose, it can be assumed that some improvements and enhancement are to be implemented in the future:

i. **Integration of Online payment gateway**
   The use of secure payment gateways like eSewa, Khalti or international payment services could facilitate real time online payment and make the user experience more convenient.
ii. **Next-Gen Reporting and Analytics**
   Sales reports, revenue analysis, customer behaviour tracking, and graphical dashboards can be added to the system to form a complete Management Information System (MIS).
iii. **Enhanced Security Measures**
   Introduction of HTTPS, better passwords, multi-factor authentication, and role-based access control would also enhance the safety of the system.
iv. **Mobile Application Support**
   It would be better to create a mobile-friendly or specific mobile application to enhance accessibility and user interaction.
v. **Cloud Deployment and Scalability**
   The implementation of the system on the scalable cloud with improved databases would enable the system to support increased traffic and volumes of transactions.

Such improvements would go a long way in increasing the functionality, usability, and sustainability of the systems.

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